```
class QuadraticMean(gpytorch.means.Mean):
    def __init__(self, batch_shape=torch.Size(), bias=True):
        super().__init__()
        self.register_parameter(
            name="second",
            parameter=torch.nn.Parameter(torch.randn(*batch_shape, 1, 1))
        )  ❶

        self.register_parameter(
            name="first",
            parameter=torch.nn.Parameter(torch.randn(*batch_shape, 1, 1))
        )  ❷

        if bias:
            self.register_parameter(
                name="bias",
                parameter=torch.nn.Parameter(torch.randn(*batch_shape, 1))
            )  ❸
        else:
            self.bias = None
```

❶ Second-order coefficient.

❷ First-order coefficient.

❸ Bias.

We sequentially call `register_parameter()` for register the second-order coefficient, the first-order coefficient, and the bias. As we don't have a good idea of what values these coefficients should take, we simply initialize them randomly using `torch.randn()`.

NOTE **We need to register these parameters as instances of the `torch.nn.Parameter` class, which allows their values to be adjusted (trained) during gradient descent.**

For the `forward()` method, we need to define how our mean function should process an input. As we have said, a quadratic function is in the form of $ax^2 + bx + c$, where *a*, *b*, and *c* are the second-order coefficient, first-order coefficient, and the bias, respectively. So we only need to implement that logic like so:

```
class QuadraticMean(gpytorch.means.Mean):
    def __init__(self, train_x, train_y, likelihood):
        ...  ❶

    def forward(self, x):
        res = x.pow(2).matmul(self.second).squeeze(-1) \
            + x.matmul(self.first).squeeze(-1)  ❷
        if self.bias is not None:
            res = res + self.bias
        return res
```

❶ Omitted.

❷ Formula of a quadratic function in Figure 3.9.

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